

Classification problem - Whether a user purchased based on Social Media Ads

Random Forest Algorithm (RF)

Confusion Matrix

```
[74,  9]
[ 2, 35]
```

Classification Report:

	precision	recall	f1-score	support
0	0.97	0.89	0.93	83
1	0.80	0.95	0.86	37
accuracy			0.91	120
macro avg	0.88	0.92	0.90	120
weighted avg	0.92	0.91	0.91	120

Decision Tree Algorithm (DT)

Confusion Matrix

```
[71 12]
[ 4 33]
```

Classification Report:

	precision	recall	f1-score	support
0	0.95	0.86	0.90	83
1	0.73	0.89	0.80	37
accuracy			0.87	120
macro avg	0.84	0.87	0.85	120
weighted avg	0.88	0.87	0.87	120

SVM Algorithm (SVM)

Confusion Matrix

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[80  3]
[22 15]
```

Classification Report:

	precision	recall	f1-score	support
0	0.78	0.96	0.86	83
1	0.83	0.41	0.55	37
accuracy			0.79	120
macro avg	0.81	0.68	0.71	120
weighted avg	0.80	0.79	0.77	120

0: Did Not Purchase

1: Purchased

Questions:

1. How often does the model correctly identify people who did not purchase?

Ans:- Recall (0) - **(RF) 0.89 (DT) 0.86 (SVM) 0.96**

Out of all actual non-buyers, **(RF) 89% (DT) 86% (SVM) 96%** were correctly predicted as not purchasing.

2. How often does the model correctly identify actual buyers?

Ans:- Recall (1) - **(RF) 0.95 (DT) 0.89 (SVM) 0.41**

It correctly identifies **(RF) 95% (DT) 89% (SVM) 41%** of the actual buyers.

3. When the model says someone didn't purchase, how often is it right?

Ans:- Precision (0) - **(RF) 0.97 (DT) 0.95 (SVM) 0.78**

The model is **(RF) 97% (DT) 95% (SVM) 78%** correct when it predicts someone didn't purchase.

4. When the model predicts someone will purchase, how often is it right?

Ans:- Precision (1) - **(RF) 0.80 (DT) 0.73 (SVM) 0.83**

The model is correct **(RF) 80% (DT) 73% (SVM) 83%** of the time when it predicts a buyer.

5. Is the model good at not missing or wrongly predicting non-buyers?

Ans:- F1-score (0) - **(RF) 0.93 (DT) 0.90 (SVM) 0.86**

The model has a high F1 score, meaning good balance for non-purchasers.

6. How well does the model handle mistakes for buyers?

Ans:- F1-score (1) - **(RF) 0.86 (DT) 0.80 (SVM) 0.55**

The model balances catching buyers with minimizing false alarms. But with SVM, The model is missing many real buyers (low recall)

7. What percentage of all predictions are correct?

Ans:- Accuracy - **(RF) 0.91 (DT) 0.87 (SVM) 0.79**

The model is correct **(RF) 91% (DT) 87% (SVM) 79%** of the time overall

8. How balanced is the performance across both classes?

Ans:- Macro avg (f1-score) - **(RF) 0.90 (DT) 0.85 (SVM) 0.71**

On average, across classes, F1-score is **(RF) 90% (DT) 85% (SVM) 71%**— showing good balance

9. How does the model perform considering class imbalance?

Ans:- Weighted avg (f1-score) - **(RF) 0.91 (DT) 0.87 (SVM) 0.77**

Weighted performance is also high **(RF) 91% (DT) 87% (SVM) 77%**, indicating good real-world robustness

10. Does the model perform equally well across both classes?

Ans:- Support **(0=83, 1=37)** - Imbalanced

Class 0 (non-purchasers) is more common, but the model still performs well on the minority Class